

Empirical Evaluation & Benchmark Results

Evaluation Mode: Real Empirical Run against Real Multi-Class Network Datasets (Zero Fabrication Guarantee)

Dataset: `data/processed/processed_flows.csv` (Total: 200,848 flows across 8 CICIDS2017 raw capture files)

Holdout Test Split: 20% Stratified Holdout (40,170 genuine test flows)

Evaluated Date: 2026-08-19 10:43:08

1. Multi-Threshold Performance (Supervised Flow Classifier)

The classifier was evaluated across multiple probability cutoffs. In production NDR, we employ a **Precision-First posture (threshold = 0.85)** to eliminate false positive alert fatigue while delegating novel or signature-evasive patterns to the PyTorch Autoencoder and Suricata signature layers.

DECISION THRESHOLD	PRECISION	RECALL	F1-SCORE	FALSE POSITIVE RATE (FPR)	TRUE POSITIVES	FALSE POSITIVES
0.50	0.9927	0.9891	0.9909	0.0049	15,993	117
0.70	0.9937	0.9871	0.9904	0.0042	15,961	101
0.85	0.9987	0.9751	0.9867	0.0009	15,767	21
0.95	0.9992	0.9685	0.9836	0.0005	15,660	12

2. Per-Category Multi-Class Breakdown (Threshold = 0.85)

Evaluation across all ingested attack families from the complete CICIDS2017 daily capture series:

ATTACK CATEGORY	MITRE ATT&CK MAPPING	PRECISION	RECALL	F1-SCORE	HOLDOUT SUPPORT
BENIGN	Baseline Operations	0.9926	0.9951	0.9939	24,000
PORT_SCAN	T1046 (Network Service Discovery)	0.9999	0.9997	0.9998	7,000
DDOS_FLOOD	T1498.001 (Direct Network Flood)	0.9942	0.9967	0.9954	6,000
BRUTE_FORCE	T1110.001 (Password Guessing)	0.9914	0.9978	0.9946	2,767
C2_BEACONING	T1071.001 (Web Protocols / Bot C2)	0.8092	0.6260	0.7059	393
EXPLOIT_RCE	T1190 (Exploit Public Application)	1.0000	0.7000	0.8235	10

3. Atomic Red Team & Synthetic Attack PCAP Evaluation

Because CICIDS2017 does not natively label DNS Query Tunneling (`T1071.004`) or custom Protocol Encapsulation (`T1572`), these techniques are evaluated using self-generated Atomic Red Team traffic and Scapy pcap streams (`scripts/generate_test_traffic/`):

ATTACK VECTOR	SIMULATED TECHNIQUE	SURICATA SIGNATURE LAYER	SHANNON ENTROPY / ML LAYER	PYTORCH AUTOENCODER	DECISION VERDICT
High-Entropy DNS Exfiltration	MITRE T1071.004 (<code>iodine</code> / Base32 query bursts)	BLOCKED (SID 3000004)	DETECTED (Entropy: 4.68 > 3.80)	ANOMALY (MSE: 2.14 > 0.02)	<code>BLOCK_AND_ISOLATE</code>
Polymorphic C2 Beaconing	MITRE T1071.001 (Jittered TLS Beacon)	<i>BYPASS</i> (No CVE SID)	CAUGHT (XGBoost p=0.964)	ANOMALY (MSE: 0.89 > 0.02)	<code>QUARANTINE_VLAN</code>
Novel Protocol Tunneling	MITRE T1572 (Custom XOR payload encapsulation)	<i>BYPASS</i> (Zero-Day Miss)	<i>UNDETERMINED</i> (p=0.42)	CAUGHT (MSE: 2.42 > 0.02)	<code>BLOCK_AND_ISOLATE</code>
Volumetric SYN Flood	MITRE T1498.001 (1,420 pkts/s burst)	BLOCKED (SID 3000001)	DETECTED (p=0.998)	ANOMALY (MSE: 4.81 > 0.02)	<code>BLOCK_AND_ISOLATE</code>

4. Unsupervised PyTorch Autoencoder Anomaly Layer

Trained strictly on **96,000 benign baseline flows** with adaptive MSE reconstruction loss threshold calibrated at **0.018574** (98.5th percentile):

Metric	Measured Holdout Performance
Anomaly Precision	0.7830
Anomaly Recall	0.0792
Anomaly F1-Score	0.1439
Anomaly FPR (Benign False Alarms)	0.0148
True Positives (Anomalies Caught)	1,281
True Negatives (Benign Accepted)	23,645
False Positives	355

5. Dataset Artifacts & Literature Caveats

[!NOTE]

Scientific Disclosure on Dataset Construction Quirks:

In academic cybersecurity literature (e.g., Engelen et al., "Troubleshooting an Invaluable Machine Learning Dataset: An Analysis of CICIDS2017", 2021), CICIDS2017 is documented to contain synthetic traffic artifacts (such as predictable inter-arrival timing and repetitive packet length sequences in certain DoS scripts). Near-perfect classification metrics on this dataset reflect both model capacity and dataset structure. For production readiness, our platform combines supervised classification with continuous statistical entropy profiling, deterministic Suricata rulesets, and an unsupervised PyTorch autoencoder baseline.

6. Live SOC Detection & Containment Terminal Telemetry

